

Laser Scanning

- technique to collect explicit 3D coordinates
 - point clouds
 - for terrain / objects
- laser range finder + scanning mechanism
- dense 3D data
- mapping, modeling, analysis

history

- light for ranging → ruby crystal light beam
→ laser range finder

uses

- atmospheric, satellite, lunar

terrestrial: total station (reflectorless)

- tunnels, shafts, bridges

Capability

- progressed from single distance to 3D scanning
- improved precision and resolution, measurement rates

Impact

- incorporate with mapping and GIS
- accurate digital model (link virtual → reality)

Limitation

- mobility, occlusion

ALS

- single profile (nadir only), phase / pulse based
 - limited coverage, elevation ✓

Laser profiler

- collects nadir only
- wide area mapping → time consuming, inefficient

1990s scanning mechanism (mirrors)

- Air LiDAR: a broad swath of points
- direct georeference: GPS, IMU
- accurate ground point calculation
- maturity: commercially available
 - surveying, forestry, urban mapping
 - examples: NASA's ATM, Optech, Leica, Riegl
- airborne → spaceborne (use satellites)
 - elevation ✓, area scanning limited

LASER

- light amplification by the stimulated emission of radiation
- specific f and λ
- coherent beam of photons
- same λ , all in phase, same polarization
- narrow spectral domain ($< 1 \text{ nm}$)
- focused on a tight spot
- narrow spot over great distances ($< 20 \text{ m}$, $d > 100 \text{ m}$)

Laser components

① Gain medium

- material excited to produce light
- solid state (bulk): Nd:YAG
- gas: HeNe, CO₂

② Energy source

- pumping: start up / continue laser
- high intensity discharge lamp / laser diode

③ Optical resonator

- 1 fully reflective + 1 semi reflective (output coupler)

Two level pumping

- pumping, electron gets excited
- spontaneous emission: wrong direction → lost, right direction → back
- stimulated emission: by photons of the right direction

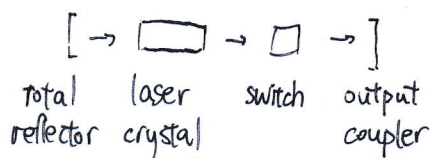
Nd:YAG peak emission: 1064 nm, highest

Laser cavity

- feedback loop, mirror to mirror
- optical path length is $\frac{1}{2} \cdot m$, accounted for refractive index

Laser pulse

- Q-switch (example of pulsed lasers)
 - delays the release of energy until very high power
- CW (continuous wave): low intensity
 - v. high intensity, penetration
- Gaussian switch
 - easy to control
 - higher intensity than CW



Timed pulse method

- not CW
- uses a short time window as pulse
- emitted pulse: from laser
- returned pulse: "echo", reflection from object
- regular scan: record return only
- high end: emitted + returned

Laser Pulse Characteristics

① pulse repetition frequency (PRF)

- pulses/s
- period: T , $PRF = 1/T$
- old: $5 \sim 25$ kHz, modern: 100 kHz ~ 2 MHz
- higher PRF:
 - pulse shape not well defined
 - $\uparrow$ signal to noise ratio
 - $\uparrow$ point density
 - $\downarrow$ energy of pulse
- lower PRF:
 - $\uparrow$ distance
 - well defined shape

② pulse duration and range resolution

- pulse width/duration affects distinguishing closely spaced targets $\Lambda \Lambda \cap$
- shorter pulses improves
- high-res lidar: < 10 ns
 - Q switching
- range resolution: ground distance ϵ
before unable to distinguish $\cap$

③ divergence / footprint size

- in mrad, size^m = divergence $\theta \cdot d$ km
- due to diffraction, energy spreads gaussian
- beam quality $\rightarrow \downarrow$ divergence
- small spot size $\rightarrow$ can use point cloud
- large spot size $\rightarrow$ invalid for small details
- common cutoff of photons: $\pm 2\sigma$, uncommon: $\pm 1\sigma$

3 Typical central wavelengths for lidar

- 1064 nm:
 - highest gain by Nd:YAG
 - most common, NIR
 - 532 nm:
 - 2x frequency, $\frac{1}{2}\lambda$, green
 - 1550 nm: NIR
 - safer $\uparrow$ reflectance for most material.
 - $\uparrow$ transmittance
 - forestry application: separate leaf, wood, soil
- green: eye safety concern, sensitive to atmosphere, water, $\uparrow$ attenuation water

Further reflectivity example

- reflectance: type of measurement
- reflectivity: intrinsic property of material
- highly reflective obj.: may saturate detectors
- low reflectivity obj.: too weak
- reflectivity $\rightarrow$ min. detectable object size
-  spot size
- object size (1%, reflects 100%) $\rightarrow$ contributes only 1%

Ranging, Profiling, Scanning

- all: based on laser ranging instrument (basic)
 - laser range finder (LRF)
- 0D: 1 point e.g. total station
- 1D: beam with moving/rotating instrument along a line e.g. satellite
- 2D: Scanning in a 2D pattern (line by line zenith + azimuth, or rotating + moving)
 - data captured is always 3D
 - only laser's FOV is 0/1/2D

Laser range finder

- Time of flight (ToF) - pulse laser $d = \frac{ct}{2}$
 $\Delta d = \frac{c \cdot \Delta t}{2}$
- $\uparrow$ pulse rate vs unambiguous range
 - lidar must allow range finish round trip, $t_{\text{pulse}} < t$
- receiver sensitivity vs ambient light
 - solar noise or signal? solar signal continuous to sensor
- hardware cost and complexity
 - innovation: $\downarrow$ cost and size of lidar. best still expensive
- CW - phase shift ranging
 - amplitude-modulated, measures phase shift emit vs received
 - $d = (M\lambda + \Delta\lambda)/2$
 - more accurate than ToF sensors
 - needs specific sensors. high pulse energy

ToF

- range: long to v. long
- accuracy: moderate
- speed: high
- in noise or ambient light: affected
- complexity: fast timers
- cost: high
- precision: v. precise timing
- examples: meas. drones, satellites

CW

- range: short (< 60 m)
- accuracy: high
- speed: high
- in noise or ambient light: less affected
- complexity: phase detection electronics
- cost: low
- precision: phase meas. in degrees
- examples: phones

Laser Profiling - terrestrial

- 1D line $\rightarrow$ maps 2D 
- 2D vertical profile / cross section

Laser scanning

- map topographic features of area in detail
- TLS: azimuth scan $\times$ series of terrain profiles
- ALS: cross track scan
 - profile is line of swath width
- SLS does not exist

ALS mirrors

- oscillating mirrors 
- rotating polygon mirror 
- rotating mirror  (or grid)
- rotating wedge prism 
 - or palmer scan

Ranging

- OD
- beam fixed
- total station

Profiling

- 1D
- platform motion
- airborne altimeter

Scanning

- 2D
- 2 axis motion
- TLS, MLS, ALS
- dense topographic model

Lidar system

- 3D laser scanning system
- beam steering, optics
- emitter, receiver
- GNSS, IMU (mobile)

FOV

- 2-4 times larger than divergence
- ensures capturing beam div.

Angular steps / resolution

- smallest angle to the next meas. direction
- e.g. 0.01° at 60m $\rightarrow$ 1.7cm
- too small: redundancy

Footprint size = angular resolution / divergence

 $\rightarrow$ seamless scan $\checkmark$

 $\rightarrow$ sparse $\times$

 $\rightarrow$ redundancy $\times$

Waveform

- ideal case: single return 
- multiple return (trees) 
- $\rightarrow$ temporal dimension in wave

$$R = \frac{ct}{2}, \Delta R = \frac{c\Delta t}{2}$$

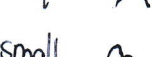
Discrete return lidar vs waveform lidar

- discrete $\rightarrow$ point cloud
- set amplitude as threshold
- does not record the shape of waveform
 - storage is cheaper
- waveform
 - rare and costly, TB level
- most common: 1-3 pulse, first, last, max intensity

Gaussian decomposition

- fit multiple gaussian profiles $\rightarrow$ least squares

Interactions between emitted pulse and target

1. $\perp$ to pulse direction 
2. oblique to pulse direction 
3. a cluster of targets 
 - very small footprint size

Terminology

- temporal bin size 
 - quantization of internal timer
- range resolution 
 - smallest separation of beams
- range accuracy and precision
 - meas. distance Δd and σd

Ranked by sampling rate and resolution

- TLS $\rightarrow$ HLS / backpack $\rightarrow$ MLS
- UAV $\rightarrow$ aircraft $\rightarrow$ satellites

Single photon lidar

- detects probability of signal
 - removes noise from EM waves
 - new tech in NASA satellites
- classified by targets
- atmospheric
 - land
 - bathymetric

Solid angle

- $\Delta\Omega = \frac{A}{r^2} \approx \pi(\frac{\beta}{2})^2$, $\beta = \text{FOV}$
- small: high res, small coverage
- large: wide coverage, low res

Pulse energy vs Power

- Pulsed lidar
- CW

Overlap factor

$[0, 1]$ 

Target area $A_t(R)$

- obj. < FP: illuminated area target area back scatter area constant energy $\frac{1}{R^4}$
- obj. > FP: FP area R^2 $\frac{1}{R^2}$

small common in: satellites.

big common in: TLS, HLS

Lidar equation Assumptions

- Lambertian surface
- $\perp$ pulse direction
- uniform energy divergence

3D Point cloud

- 3D points w/ RGB + intensity
- unstructured unlike 2D image
- format: las
 - 1.2, 1.3: point cloud data fields
 - 1.4: waveform
- reasons:
 - instant capture 3D scene
 - geometric measurements
- use: 2D: topographic
3D: as built records
 - traffic surveillance, auto mobiles

Defects

- RGB / point cloud / strip misalignment
- occlusion
- dark obj., high surface moisture
- swath gap
- Speckle (glass / water)

Point Cloud software

- generators: cyclone, DJI Terra
- processing and production: LASTools, 3DMas
- application: ArcGIS, Revit

Pre processing

- cloud slicing: partition point cloud
 - e.g. 1000 grids
- downsampling: memory vs details
- noise / outlier removal: req. void filling
 - may remove features
- coordinate transformation: void filling

Pre-processing

- normal estimation: align a surface
- plane segmentation / extraction: e.g. ceiling, wall, floor
- cylindrical estimation

Point Cloud classification / semantic labeling

- application purpose known
- with specific meaning
- classify point clouds
- self design
- e.g. land cover: tree, grass, road, soil, building
- BIM: wall, ceiling, floor, window

Future development

- AI semantic segmentation
- auto map feature extraction, vectorization

Classification

Segmentation

- single label, object recognition
- categorical label
- table 1, table 2
- all tables

Segmentation

Semantic

Instance

Panoptic

- labels every point based on semantic class

- distinguish obj in same class

- Semantic + instance

points class 1 class 2 class 3 obj 1 obj 2 obj 3 class 1 class 2 class 3 obj 1 obj 2 obj 3

- purpose
 - convert xyz to actionable intelligence
 - thematic mapping and extract geospatial statistics

e.g. avg height of trees.

sand for reclamation

- understand 3D scene

- no need to measure close up $\rightarrow$ safety
- AI to recognize obj $\rightarrow$ efficiency

oscillating mirror vs rotating polygon

- ⊕ maintain swath width ⊖ going over mountains, cannot maintain
- ⊖ uneven intensity ⊕ even intensity

- Limitations:
- data volume ↑
 - data defects
 - unstructured, sparsity
 - cross sensor, modal
 - domain shift in AI (AI for place A $\neq$ place B)

Manual Segmentation

- trace polygons manually
- steps: denoise $\rightarrow$ slicing $\rightarrow$ downsampling
- slice the cloud in 3D, segment out selection $\rightarrow$ new class

Semi-supervised Segmentation (3D Mas)

- ① one-click and segmentation
 - growing algorithm
- ② project 3D pt cloud onto 2D, run segmentation model
 - e.g. Meta SAM, can ③ $\rightarrow$ ① $\uparrow$ efficiency
- ③ highest accuracy
 - act as ground truth / training data $\rightarrow$ AI

Unsupervised clustering

- partition itself w/o prior knowledge
- derive features (e.g. surface normal)
- run clustering algorithm (K-mean, DBscan)
- may produce: wall + floor + ceiling based on surface normal

Supervised learning

- trainer: decides parameters, datasets
- well labeled data: x.y.z, R.G.B.I. normal
- feature selection
 - dimension reduction
 - feature extraction
- model training (NN, random forest, maximum likelihood)
 - NN: loss function, goal min loss
- prediction: predict labels, check accuracy

End to end deep learning

- raw point clouds
- 80% of ground truth as training data
- local feature encoder: learns own features
 - 3×3 , 5×5
- global context aggregation: understand bigger picture $\rightarrow$ global feature vector
- decoder: vector $\rightarrow$ local features
- loss function

Cross modal (2D to 3D) projection

- 2D image segmentation + EO params
 - $\hookrightarrow$ 3D pt cloud
- ⊕ - fast, rough positioning and visualization
 - off the shelf vision foundation models (YOLO)
- ⊖ - boundary fuzzy
 - inaccurate project due to occlusion

Improving segmentation

- first do manual, supervised
- high quality training data
- remove noise / outliers / artifacts
- normalization / correction $\rightarrow$ homogeneous intensity
- extract representative features (e.g. H.I.I., surface normal)
- multi scale grouping (e.g. fence vs fire hydrant)
- domain adaptation: src HK $\rightarrow$ target toronto

Segmentation helps

- distinguish obj from pts
- give semantic meaning to obj.
- produce by-products: map, BIM
- actionable intelligence

Major components

- all LS: laser rangefinder, scanner. (camera)
- MLS / ALS: GNSS, IMU
 - active - compute precise vector
 - passive - RGB values
- mirror - all: rotating mirror / prism
 - ALS: oscillating / nutating mirror
- TLS: rotating body

IMU

- measures platform movement and rotation
- accelerometer, gyroscope (xyz, $\omega \phi \kappa$)
- e.g. in a tunnel, use acc to estimate

GNSS

- aircraft single antenna - affected by: I, T, m, e
- data is post processed

Radiometric quantities

- basic: Gaussian profile
- radiometry
- reflectance, deviation

Storage formats

- ASCII (txt, csv), PTX, las, HDF
- example: x y z I C
- +ve: intuitive, easy, compatible with software
- -ve: large, X header/metadata
(coordinate system, origin, units)
- PTX: ascii w/ headers
 - 10 lines: column, row, sct, x, y, z, Rmatrix, t(1,2,3)
 - followed by: x, y, z I rgb
 - special case: all zero: emit w/o return

LAS Format (ASPRS .las)

- industry standard binary format
- small, public, open
- header: point data record format
 - xs, ys, zs, Δx, Δy, Δz
 - start of waveform data packet record
 - $x(y/z) = (X_{record} \cdot s) + \Delta x$
- minimum: x, y, z
- GPS_t, rgb, I, num, ret index, C
 scan direction flag, edge of flight line

Intensity

- varies among companies, devices
- radiometric quantity
- can be absolute or dB

GPS Time

- easy to segment
- tell platform trajectory, scan mode
- interpolate pos of platform.
- $\Omega = [P_x - GPS_x, P_y - GPS_y, P_z - GPS_z]$

Scan angle

- zenith angle with sign (left, right)
- from GPS, IMU, integer, in m frame
- $\theta = \arccos(\text{norm } \Omega_z)$

Multi return ALS over forest

- PSM (surface), DTM, CHM

Waveform data packet records

- each ns, 1 record (integer)
- $x(y/z) = X_0 + t_0 \cdot dx$
- waveform location: $x(y/z) = X_p + L \cdot dx$

Extra Byte (Riegl)

- end of standard point record
- GPS_tσ, classification probability, waveform derived metrics.
- α-line: full waveform, pulse width, offline
- V-line: deviation, online

Points from waveform lidar

- Gaussian decomposition: fit waveform, multi-modal
- derivative $\rightarrow \frac{0}{-} \frac{0}{-} \dots$ zero crossing peaks
- for 1 or oblique: G. profile: $\hat{p} = e^{-\frac{(t-t_i)^2}{2\sigma^2}}$
- FWHM full width $\frac{1}{2} \max A = 2.35 \cdot \sigma$

Derived radiometric quantity

- integrated amplitude per return 
 - return energy
 - $E = \sqrt{2\pi} \cdot \hat{p} \cdot \sigma$
- E is more stable than $\hat{p}_i$ (height) to derive RQ.
- proportional to radiance
↳ lambertian surface

LIDAR equation

$$E_{\text{recv}} = E_0 \cdot \underset{\substack{\text{system} \\ \text{efficiency}}}{\eta_{\text{sys}}} \cdot \underset{\substack{\text{overlap} \\ \text{area}}}{G} \cdot \underset{\substack{\text{recv} \\ \text{area}}}{\frac{A_{\text{recv}}}{4\pi R^2}} \cdot \underset{\substack{\text{backscatter} \\ \text{area}}}{\frac{\Sigma R}{R^2}} \cdot \underset{\substack{\text{atmosphere}}}{T_{\text{atm}}}^2$$

$$E' = E \cdot R^2, E' > E \text{ when deriving RQ.}$$

- more correlated to reflectivity

power W (J/s) = radiant flux Surface irradiance

radiant intensity W/sr $\phi(\text{power})$ $I = \frac{\phi}{A}$ (+)

irradiance W/m²

radiance W/m²/sr $\frac{\phi}{A} \cdot \cos \theta$ (oblique)

Radiance / brightness

- to camera/eye
- invariant with distance

$$L = \frac{d^2 P}{d(A \cdot \cos \theta) \cdot d\Omega} \quad \text{solid angle}$$

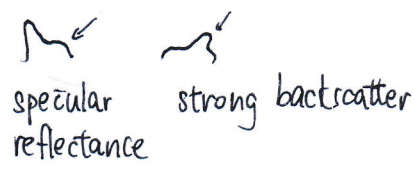
Lambertian surface

$$\Phi_{power} = L \cdot A \cdot \cos \theta \cdot \Omega$$

$$L = \rho \cdot I \cdot \frac{1}{\pi}$$

surface reflectivity

Non lambertian



- lidar ≠ sunlight
- lidar sensor, detector moves together

$$\text{lambertian: oblique } E = I E \cdot \cos \theta$$

Radiometric sensor output vs Radiance

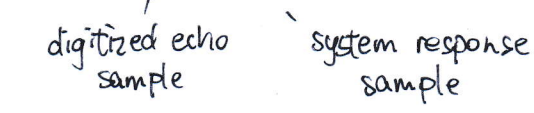
- camera: L over each pixel iFOV
- LIDAR: integrated amplitude E
- $E \propto L$
- time resolved return

Calibration

- tarp: lambertian surface, known ρ reflectivity
- calculate distance corrected E'
- $E'_{corr} = E' \cdot l \cdot n = E' \cdot \cos \theta$

deviation of echo pulse shape

$$\delta = \sum s_i - p_i$$



Point filter by pulse shape deviation

- 2 profile too close → merge together
- large deviation
- ghost points

Intensity

- vaguely defined
- peak amplitude (simplest)
- integrated amplitude (most correlated w/ RQ)
- distance corrected integrated amplitude

Irradiance vs radiance (target lambertian)

- sun same I, L
- near (light bulb, laser)
- $R \rightarrow A, I, L$
- $zR \rightarrow 4A, \frac{I}{4}, \frac{L}{4}$

$$RQ = 10 \left(-A_{low} + \left[\frac{DN}{0.5535} \cdot (A_{up} - A_{low}) \right] \right) / 10$$

upper lower bound dB

Equations

$$\text{ToF } d = \frac{ct}{2}, \Delta t = \frac{c\Delta t}{2}$$

$$\text{waveform bin resolution } \Delta R = \frac{c\Delta t}{2}$$

$$\text{CW: } \Phi_r = \frac{4\pi f_m d}{c}$$

$$\text{modulation wavelength } \lambda = \frac{c}{f}$$

$$\text{distance from phase difference } d = \frac{M\lambda + \Delta\lambda}{2}$$

$$\Delta\lambda = \frac{\Delta\phi}{360^\circ} \lambda_m$$

Laser pulse & beam

$$\text{footprint diameter } D = \beta \cdot R \quad (\beta \text{ in rad})$$

$$\text{pulse repetition frequency } PRF = \frac{1}{T}$$

Gaussian waveform decomposition

$$P = \sum \hat{P}_i \exp \left(\frac{-(t-t_i)^2}{2\sigma_i^2} \right)$$

$$\text{integrated amplitude } E = \sqrt{2\pi} \cdot \hat{P}_i \cdot \sigma$$

$$FWHM = 2.35\sigma$$

$$\text{distance corrected int. amp. } E' = E \cdot R^2$$

$$\text{back scatter cross section } \Sigma = 4\rho A$$

$$\text{- large: } E \propto \frac{1}{R^2}, \text{ small: } E \propto \frac{1}{R^4}$$

$$I = \frac{\Phi}{A}, I = I_p \cos \theta, L = \frac{\rho I}{\pi} = \frac{P}{A \cos \theta \cdot \Omega} = \rho \frac{I_p \cos \theta}{\pi}$$

$$E = E_p \cdot \cos \theta, E' = E_p' \cdot \cos \theta$$

$$\text{apparent reflectance } \rho_a = \frac{E'}{E_{perp}}$$

Vector Math

$$(\theta_{zenith} \varphi_{azimuth}) \quad e_x = \sin \theta \sin \varphi, e_y = \sin \theta \cos \varphi, e_z = \cos \theta$$

$$P_{global} = R \cdot \hat{R}_{lidar} + T$$

$$\text{surface normal } N = V_{31} \times V_{32} \quad \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\cos \theta = l \cdot n \quad (\text{rev. pulse direction} \times \text{surface normal})$$

$$\text{Solid angle circular beam } \Delta\Omega = 2\pi (1 - \cos \frac{\beta}{2})$$

$$\text{- hemisphere } \Omega = 2\pi \text{ sr} \approx \pi \left(\frac{\beta}{2} \right)^2$$

$$N_{cls} = \frac{\Delta\theta}{\delta\theta} \cdot \frac{\Delta\varphi}{\delta\varphi}$$

$$\Delta\Omega = (\cos \theta_0 - \cos \theta_1) \Delta\varphi$$

$$\theta = \arccos((\cos \theta_0 + \cos \theta_1) / 2)$$

$$\Delta\theta = \sin \theta \Delta\varphi$$

$$R = \begin{bmatrix} 0 & -\beta & 0 \\ \beta & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad PD = \frac{N}{A} = \frac{PRF}{v \cdot W}$$

$$W = 2H \tan \frac{\beta}{2}$$